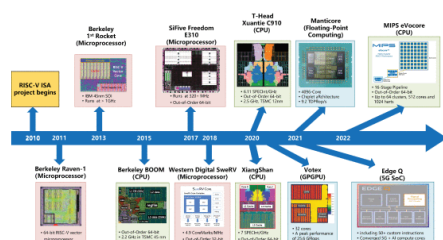


# Week 1 RISC-V Overview

## Topic Overview



### Review RISC-V ISA and Performance Evaluation.

#### Description

This module introduces the basic concepts of the Instruction Set Architecture (ISA), the minimum number of instructions directly implemented in hardware. We will talk about RISC-V ISA basics, how to use it to develop functions, and recursion in assembly language. Learning Outcomes

After the successful completion of this learning unit, you will be able to....:

1. Define the concept of ISA specifically RISC\_V 32bit ISA
2. Compare different types of caches and their influence on the CPU execution time
3. Develop programs in RISC\_V ISA to execute recursive function
4. To compute and measure the performance of a computer system.

The above Learning Objectives above align with the Course Learning Objective (list here):

## This Week's Tasks

### 1. Materials:

#### 1. Class Slides:

1. Slides for [1.Intro-333.pdf](https://canvas.calpoly.edu/courses/186931/files/21847954/download?wrap=1) (<https://canvas.calpoly.edu/courses/186931/files/21847954/download?wrap=1>) [↓](https://canvas.calpoly.edu/courses/186931/files/21847954/download?download_frd=1) ([https://canvas.calpoly.edu/courses/186931/files/21847954/download?download\\_frd=1](https://canvas.calpoly.edu/courses/186931/files/21847954/download?download_frd=1)) (<https://canvas.calpoly.edu/courses/186931/files/21847908/download?wrap=1>)
2. Slides for RISC-V assembly [2.RiscV-ISA-1.pdf](https://canvas.calpoly.edu/courses/186931/files/21847956/download?wrap=1). (<https://canvas.calpoly.edu/courses/186931/files/21847956/download?wrap=1>) [↓](https://canvas.calpoly.edu/courses/186931/files/21847956/download?download_frd=1) ([https://canvas.calpoly.edu/courses/186931/files/21847956/download?download\\_frd=1](https://canvas.calpoly.edu/courses/186931/files/21847956/download?download_frd=1)) [Function example\(RISC V assembly\)](https://docs.google.com/document/d/1WoRUD7d2sZNMsfPabKKD1T5z0cU5B3Pn/edit?usp=sharing&oid=105811798489836605177&rtopof=true&sd=true) [↗](https://docs.google.com/document/d/1WoRUD7d2sZNMsfPabKKD1T5z0cU5B3Pn/edit?usp=sharing&oid=105811798489836605177&rtopof=true&sd=true) (<https://docs.google.com/document/d/1WoRUD7d2sZNMsfPabKKD1T5z0cU5B3Pn/edit?usp=sharing&oid=105811798489836605177&rtopof=true&sd=true>) [FuncMultTwoNumbers](https://canvas.calpoly.edu/courses/186931/files/21847970?wrap=1) (<https://canvas.calpoly.edu/courses/186931/files/21847970?wrap=1>) [↓](https://canvas.calpoly.edu/courses/186931/files/21847970/download?download_frd=1) ([https://canvas.calpoly.edu/courses/186931/files/21847970/download?download\\_frd=1](https://canvas.calpoly.edu/courses/186931/files/21847970/download?download_frd=1))
3. [↗](https://docs.google.com/document/d/1WoRUD7d2sZNMsfPabKKD1T5z0cU5B3Pn/edit?usp=sharing&oid=105811798489836605177&rtopof=true&sd=true) (<https://docs.google.com/document/d/1WoRUD7d2sZNMsfPabKKD1T5z0cU5B3Pn/edit?usp=sharing&oid=105811798489836605177&rtopof=true&sd=true>) Book Chapter with more assembly code examples and more detailed explanations of the ISA [6---Architecture 2022 Digital-Design-and-Computer-Architecture.pdf](https://canvas.calpoly.edu/courses/186931/files/21847955?wrap=1) (<https://canvas.calpoly.edu/courses/186931/files/21847955?wrap=1>) [↓](https://canvas.calpoly.edu/courses/186931/files/21847955/download?download_frd=1) ([https://canvas.calpoly.edu/courses/186931/files/21847955/download?download\\_frd=1](https://canvas.calpoly.edu/courses/186931/files/21847955/download?download_frd=1))

#### 2. CPU Performance Evaluation

1. [4.Performance.pdf](https://canvas.calpoly.edu/courses/186931/files/21847948?wrap=1) (<https://canvas.calpoly.edu/courses/186931/files/21847948?wrap=1>) [↓](https://canvas.calpoly.edu/courses/186931/files/21847948/download?download_frd=1) ([https://canvas.calpoly.edu/courses/186931/files/21847948/download?download\\_frd=1](https://canvas.calpoly.edu/courses/186931/files/21847948/download?download_frd=1)) , and can use this for some [Performance practice .](https://canvas.calpoly.edu/courses/186931/pages/performance-practice-problems) (<https://canvas.calpoly.edu/courses/186931/pages/performance-practice-problems>)
2. Book Reading resources here: [COD ch 1.5-1.6.pdf](https://canvas.calpoly.edu/courses/186931/files/21847968/download?wrap=1) (<https://canvas.calpoly.edu/courses/186931/files/21847968/download?wrap=1>) [↓](https://canvas.calpoly.edu/courses/186931/files/21847968/download?download_frd=1) ([https://canvas.calpoly.edu/courses/186931/files/21847968/download?download\\_frd=1](https://canvas.calpoly.edu/courses/186931/files/21847968/download?download_frd=1))
3. Some Assembly Practice [Exercises](https://docs.google.com/document/d/1PALYzY19zqtZPCelDbSaVBAmJAYUWjcha_OMX1RoGRU/edit) [↗](https://docs.google.com/document/d/1PALYzY19zqtZPCelDbSaVBAmJAYUWjcha_OMX1RoGRU/edit) ([https://docs.google.com/document/d/1PALYzY19zqtZPCelDbSaVBAmJAYUWjcha\\_OMX1RoGRU/edit](https://docs.google.com/document/d/1PALYzY19zqtZPCelDbSaVBAmJAYUWjcha_OMX1RoGRU/edit))

## 2. Activities and Formative Assessment

1. **(Student interaction learning activity):** Introduce yourself by answering the following discussion questions; reply to two other peers adding comments, things in common, and anything constructive. [Week 1 Discussion | Introduce Yourself to the Class \(https://canvas.calpoly.edu/courses/186931/discussion\\_topics/898832?wrap=1\)](https://canvas.calpoly.edu/courses/186931/discussion_topics/898832?wrap=1)
2. **(Formative assessment):**
  1. Please complete the **Quiz: Computer Architecture Basics**. [Quiz 1 \(https://canvas.calpoly.edu/courses/186931/assignments/1612092?wrap=1\)](https://canvas.calpoly.edu/courses/186931/assignments/1612092?wrap=1)
  2. Complete the assignment [lab 1 RISC-V Matrix-Matrix Multiplication Lab \(https://canvas.calpoly.edu/courses/186931/assignments/1612100\)](https://canvas.calpoly.edu/courses/186931/assignments/1612100)
  3. Complete the assignment on performance evaluation [lab 2 Performance \(https://canvas.calpoly.edu/courses/186931/assignments/1612101\)](https://canvas.calpoly.edu/courses/186931/assignments/1612101)